EXERCISE 4: SQL JOINS, UNION, FILTERING AND AGGREGATES

Table 1: Employees

EmployeeID	FirstName	LastName	Department	Salary	City	Age
1	John	Doe	IT	70000	New York	35
2	Alice	Smith	HR	60000	Los Angeles	40
3	Bob	Johnson	Finance	75000	Toronto	45
4	David	Brown	IT	72000	London	30
5	Emma	Wilson	Sales	65000	Sydney	28
6	Michael	Clark	Finance	80000	New York	50

Table 2: Projects

ProjectID	EmployeeID	ProjectName	Budget	StartDate	Status
101	1	AI Development	100000	2024-01-10	Completed
102	2	Employee Training	50000	2024-02-15	Ongoing
103	1	Cybersecurity Audit	75000	2024-03-05	Pending
104	3	Financial Analysis	90000	2024-04-12	Ongoing
105	5	Market Expansion	65000	2024-05-20	Completed
106	6	Risk Management	80000	2024-06-10	Pending

1.
```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
FROM Employees AS A
~~LEFT~~ INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

2.
```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
FROM Employees AS A
LEFT JOIN Project AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectName	ProjectID	Budget	Status
1	John	Doe	IT	70000	AI Development	101	100000	Completed
1	John	Doe	IT	70000	Cybersecurity Audit	103	75000	Pending
2	Alice	Smith	HR	60000	Employee Training	102	50000	Ongoing
3	Bob	Johnson	Finance	75000	Financial Analysis	104	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	Market Expansion	015	65000	Completed
6	Michael	Clark	Finance	80000	Risk Management	106	80000	Pending

3. SELECT ProjectID,
ProjectName,
Budget,
Status,
EmployeeID,
FirstName,
LastName,
Department,
Salary

FROM Project AS A
RIGHT JOIN Employees AS B
ON A.EmployeeID = B.EmployeeID;

ProjectID	ProjectName	Budget	Status	EmployeeID	FirstName	LastName	Department	Salary
101	AI Development	100 000	Completed	1	John	Doe	IT	70000
103	Cybersecurity Audit	75000	Pending	1	John	Doe	IT	70000
102	Employee Training	50000	Ongoing	2	Alice	Smith	HR	60000
104	Financial Analysis	90 000	Ongoing	3	Bob	Johnson	Finance	75000
105	Market Expansion	65000	Completed	5	Emma	Wilson	Sales	65000
106	Risk Management	80000	Pending	6	Michael	Clark	Finance	80000

4.
```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
FROM Employees AS A
FULL OUTER JOIN Project AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	50000	Pending

```
ECT EmployeeID,
    FirstName,
    LastName,
    Department,
    Salary
FROM Employees
WHERE Salary > 70 000;
```

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
5	Michael	Clark	Finance	80 000

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
M Employees
ERE Department IN ('IT', 'Finance');
```

loyeeID	FirstName	LastName	Department	Salary
	John	Doe	IT	70 000
	Bob	Johnson	Finance	75000
	David	Brown	IT	72000
	Michael	Clark	Finance	80000

9. SELECT ProjectID,
ProjectName,
Budget,
Status
FROM Projects
WHERE Status IN ('Ongoing', 'Pending');

ProjectID	ProjectName	Budget	Status
102	Employee Training	50000	Ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

10. SELECT ProjectID,
ProjectName,
Budget,
Status
FROM Projects
WHERE Budget > 70 000 AND Status IN ('Ongoing', 'Pending');

ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80 000	Pending

11. SELECT EmployeeID,
FirstName,
LastName,
Department,
Salary,
City
FROM Employees
WHERE City IN ('New York', 'Toronto')
ORDER BY Salary DESC;

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80 000	New York
3	Bob	Johnson	Finance	75 000	Toronto
1	John	Doe	IT	70 000	New York

12. SELECT EmployeeID,
FirstName,
LastName,
Department,
Salary
FROM Employees
ORDER BY Salary DESC
LIMIT 3;

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000

13. SELECT Department,
SUM(Salary) AS TotalSalary
FROM Employees
GROUP BY Department
ORDER BY Salary DESC;

Department	TotalSalary
IT	142 000
Finance	155 000
Sales	65 000
HR	60 000

14. SELECT City,
AVG(Salary) AS Average Salary
FROM Employees
WHERE Salary > 65 000
GROUP BY City;

City	Average Salary
New York	75 000
Toronto	75 000
London	72 000
Sydney	65 000

15. SELECT Department,
COUNT(EmployeeID) AS EmployeeCount
FROM Employees
WHERE EmployeeCount > 1
GROUP BY Department;

Department	EmployeeCount
IT	2
Finance	2

16. SELECT Status,
COUNT(ProjectID) AS ProjectCount
FROM Projects
GROUP BY Status;

Status	ProjectCount
Completed	2
Ongoing	2
Pending	2

17. SELECT EmployeeID,
FirstName,
LastName,
SUM(Budget) AS TotalProjectBudget
FROM Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID
WHER TotalProjectBudget > 150000;

EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	175000